NAME: AMAN RAYCHAUDHURI
ROLL NO.: 2401094
BATCH: T21
BRANCH: AI&DS

EXPERIMENT – 1

Aim: Study and Implementation of Basic Git and GitHub Operations.

This Experiment maps to LO2.

LO2: To understand and perform the basic version control operations using Git and GitHub, including repository creation, cloning, file tracking, committing, pushing, pulling, branching, merging, and remote repository management.

Theory:

Git is a distributed version control system that helps developers track changes in source code and files. It enables efficient project management through repositories, commits, branches, and remote repositories, making collaboration and version management easier. In this experiment, various Git commands are performed to understand the basic operations of version control.

Commands:

No.	Command	Description
1	git init	Initializes a new Git repository in the current directory.
2	git clone <repository-url>	Creates a local copy of a remote GitHub repository.
3	git status	Displays the current status of the working directory, staged files, and untracked files.
4	git add <file>	Adds a specific file to the staging area.
5	git add .	Adds all modified and new files to the staging area.
6	git commit -m "message"	Saves the staged changes to the local repository with a commit message.
7	git log	Displays the commit history of the repository.
8	git log --oneline	Shows a compact, one-line summary of each commit.
9	git diff	Displays the differences between the working directory and the last commit.
10	git branch	Lists all local branches and highlights the current branch.
11	git branch <branch-name>	Creates a new branch.
12	git checkout <branch-name>	Switches to the specified branch.

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No.	Command	Description
13	git checkout -b <branch-name>	Creates a new branch and switches to it immediately.
14	git switch <branch-name>	Switches to an existing branch (newer alternative to checkout).
15	git merge <branch-name>	Merges the specified branch into the current branch.
16	git remote -v	Displays the remote repositories associated with the project.
17	git remote add origin <repository-url>	Connects the local repository to a remote GitHub repository.
18	git push origin <branch-name>	Uploads local commits to the specified branch on GitHub.
19	git pull origin <branch-name>	Downloads changes from GitHub and merges them into the current branch.
20	git fetch	Downloads the latest changes from the remote repository without merging them.
21	git reset --hard HEAD	Discards all uncommitted changes and resets the working directory to the last commit.
22	git stash	Temporarily saves uncommitted changes without committing them.
23	git stash pop	Restores the most recently stashed changes and removes them from the stash.
24	git rm <file>	Removes a file from both the working directory and the Git repository.
25	git config --global user.name "Your Name"	Sets the global Git username used for commits.

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```
MINGW64:/c/Apna College/Web Development/SIMON SAY GAME/GitDemo

Aman Raychaudhuri@DESKTOP-A2KO61Q MINGW64 /c/Apna College/Web Development/SIMON SAY GAME
$ git init
Initialized empty Git repository in C:/Apna College/Web Development/SIMON SAY GAME/.git/

Aman Raychaudhuri@DESKTOP-A2KO61Q MINGW64 /c/Apna College/Web Development/SIMON SAY GAME (master)
$ git add .

Aman Raychaudhuri@DESKTOP-A2KO61Q MINGW64 /c/Apna College/Web Development/SIMON SAY GAME (master)
$ git commit -m "Initial commit"
[master (root-commit) f8fa49e] Initial commit
 3 files changed, 165 insertions(+)
 create mode 100644 app.js
 create mode 100644 index.html
 create mode 100644 style.css

Aman Raychaudhuri@DESKTOP-A2KO61Q MINGW64 /c/Apna College/Web Development/SIMON SAY GAME (master)
$ git remote add origin https:github.com/amanray10/simong.git

Aman Raychaudhuri@DESKTOP-A2KO61Q MINGW64 /c/Apna College/Web Development/SIMON SAY GAME (master)
$ git push -u origin main
error: src refspec main does not match any
error: failed to push some refs to 'https:github.com/amanray10/simong.git'

Aman Raychaudhuri@DESKTOP-A2KO61Q MINGW64 /c/Apna College/Web Development/SIMON SAY GAME (master)
$ git remote add origin https://github.com/amanray10/simong.git
error: remote origin already exists.

Aman Raychaudhuri@DESKTOP-A2KO61Q MINGW64 /c/Apna College/Web Development/SIMON SAY GAME (master)
$ git push -u origin master
ssh: Could not resolve hostname https: Name or service not known
fatal: Could not read from remote repository.

Please make sure you have the correct access rights
and the repository exists.

Aman Raychaudhuri@DESKTOP-A2KO61Q MINGW64 /c/Apna College/Web Development/SIMON SAY GAME (master)
$ git remove -v
git: 'remove' is not a git command. See 'git --help'.

The most similar command is
        remote

Aman Raychaudhuri@DESKTOP-A2KO61Q MINGW64 /c/Apna College/Web Development/SIMON SAY GAME (master)
$ git remote remove origin

Aman Raychaudhuri@DESKTOP-A2KO61Q MINGW64 /c/Apna College/Web Development/SIMON SAY GAME (master)
$ git remote add origin https://github.com/amanray10/simong.git

Aman Raychaudhuri@DESKTOP-A2KO61Q MINGW64 /c/Apna College/Web Development/SIMON SAY GAME (master)
$ git remote -v
origin  https://github.com/amanray10/simong.git (fetch)
origin  https://github.com/amanray10/simong.git (push)

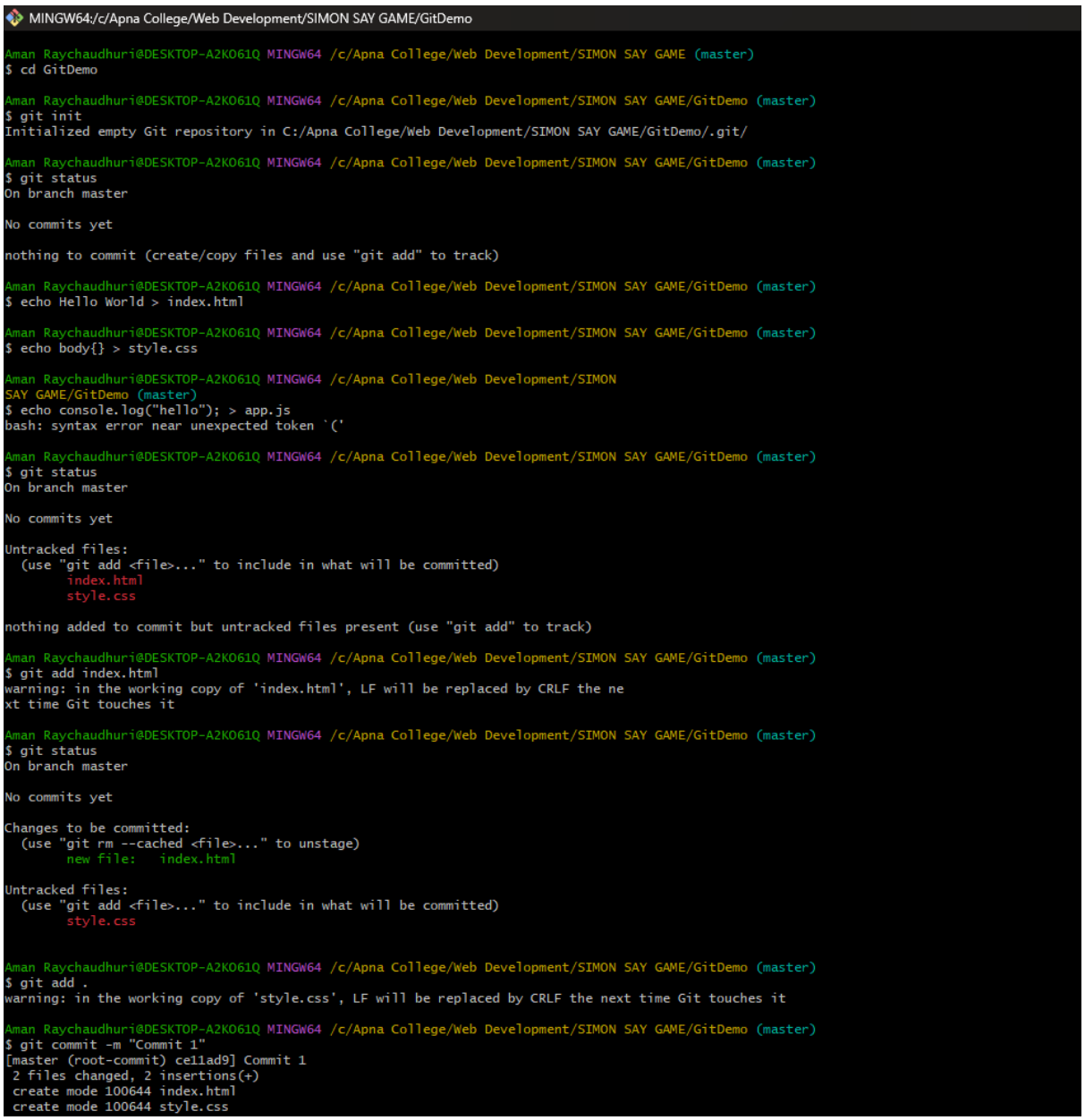
Aman Raychaudhuri@DESKTOP-A2KO61Q MINGW64 /c/Apna College/Web Development/SIMON SAY GAME (master)
$ git push -u origin master
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 1.41 KiB | 360.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/amanray10/simong.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

Aman Raychaudhuri@DESKTOP-A2KO61Q MINGW64 /c/Apna College/Web Development/SIMON SAY GAME (master)
$ mkdir GitDemo

Aman Raychaudhuri@DESKTOP-A2KO61Q MINGW64 /c/Apna College/Web Development/SIMON SAY GAME (master)
$ cd GitDemo
```

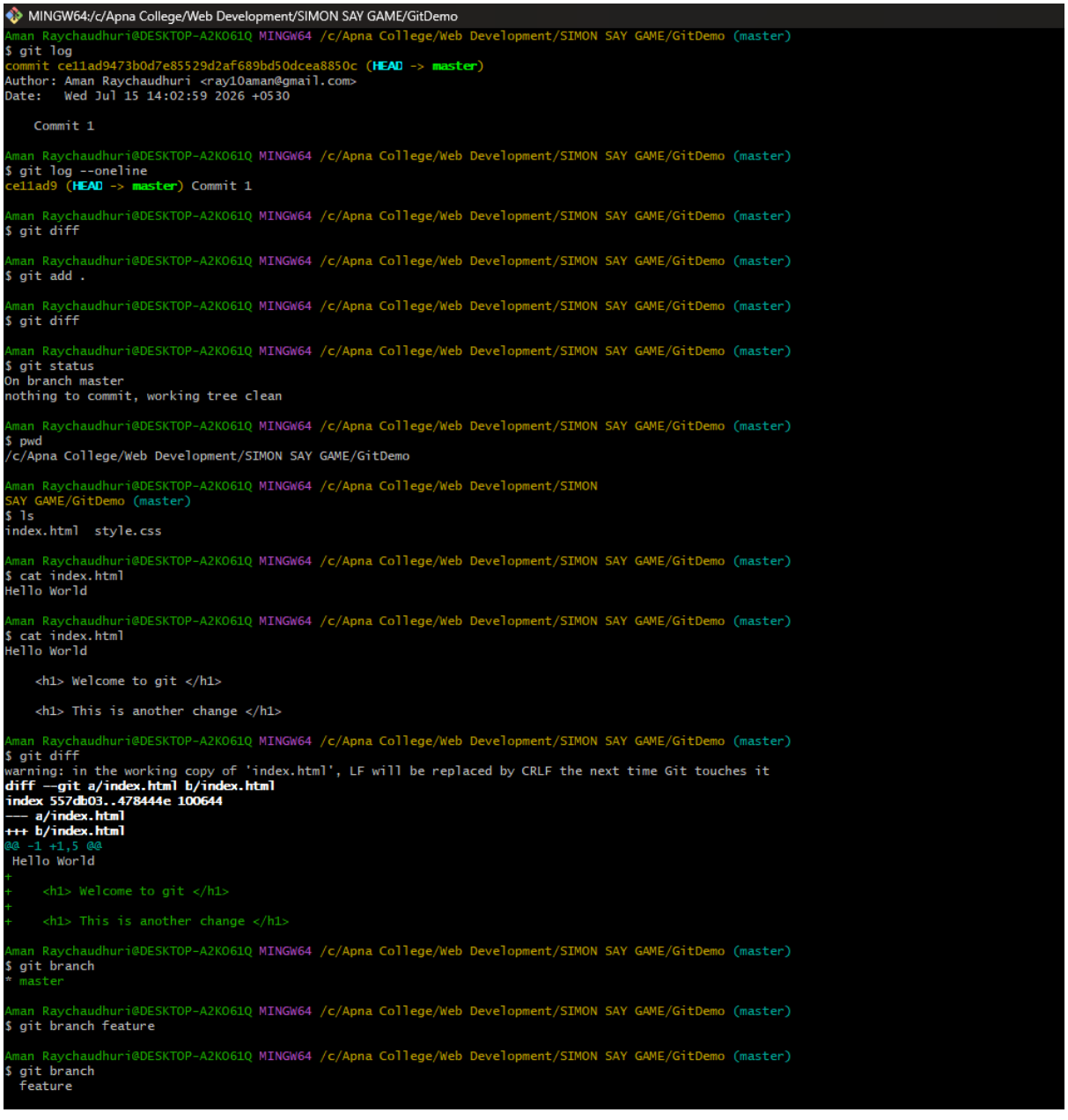
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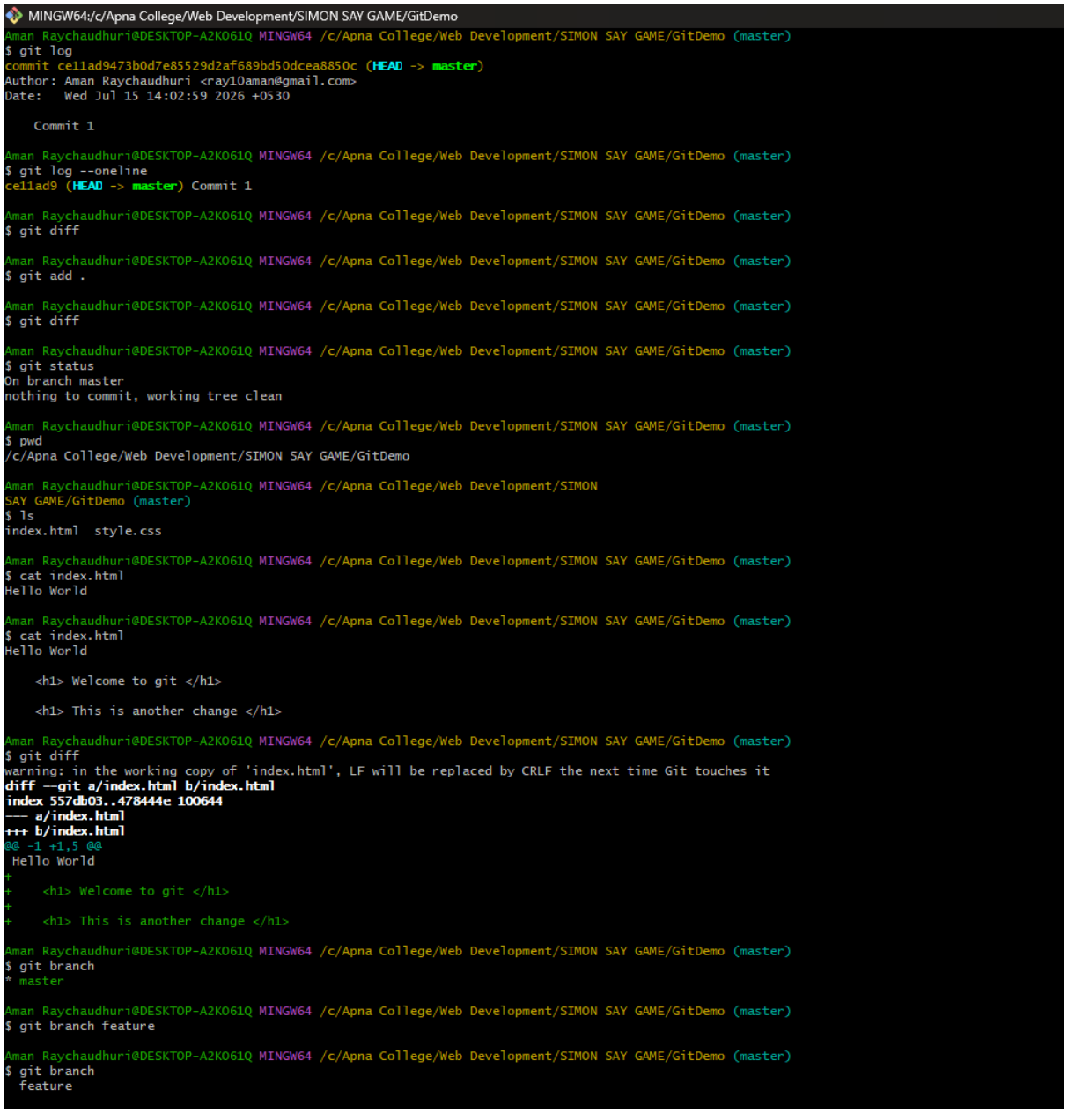
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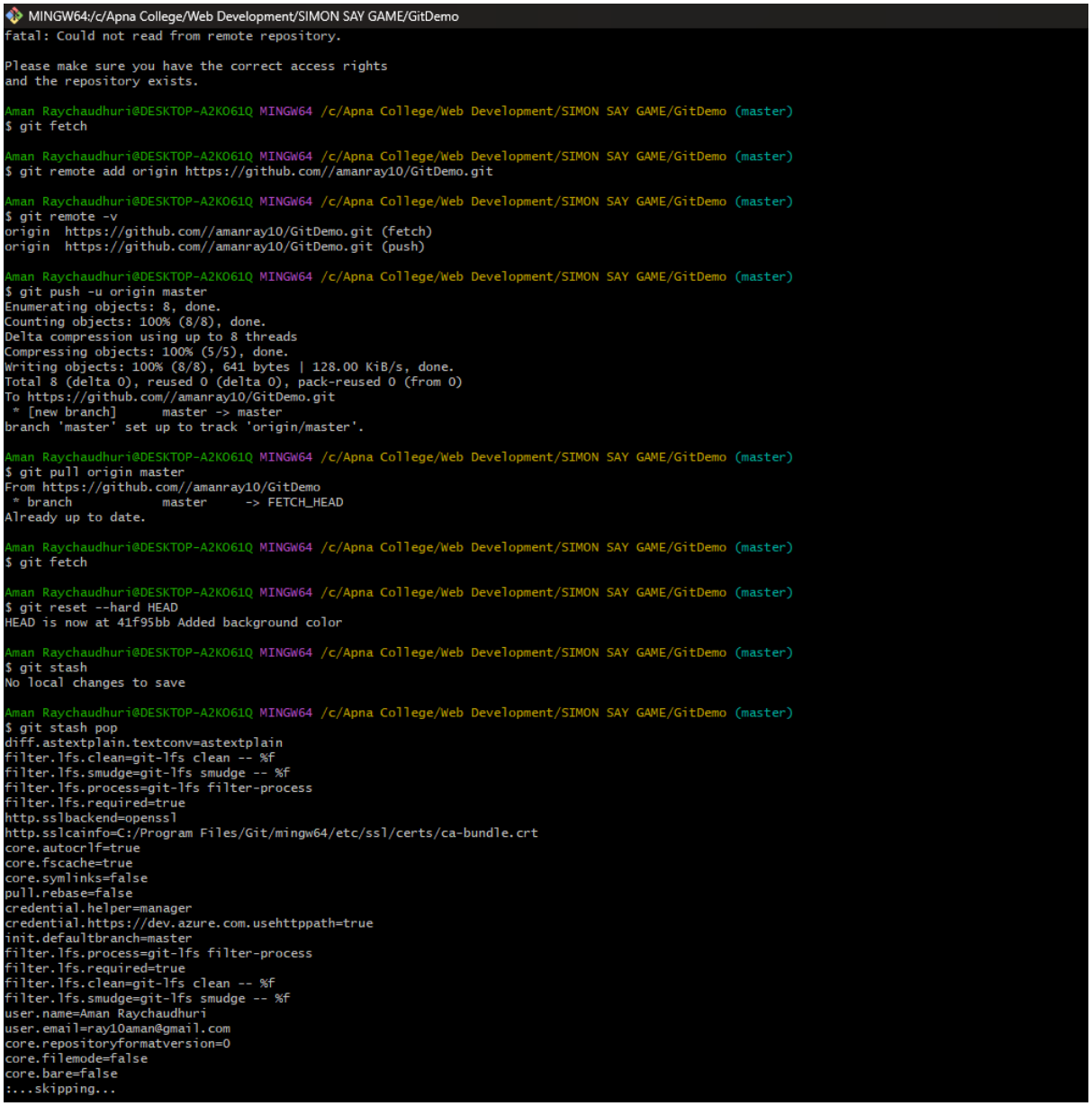
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```
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Aman Raychaudhuri@DESKTOP-A2KO61Q MINGW64 /c/Apna College/Web Development/SIMON SAY GAME/GitDemo (master)
$ git rm style.css
fatal: pathspec 'style.css' did not match any files

Aman Raychaudhuri@DESKTOP-A2KO61Q MINGW64 /c/Apna College/Web Development/SIMON SAY GAME/GitDemo (master)
$ git commit --"removed style.css"
error: unknown option `removed style.css'
usage: git commit [-a | --interactive | --patch] [-s] [-v] [-u[<mode>]] [--amend]
                  [--dry-run] [(-c | -C | --squash) <commit> | --fixup [(amend|reword):]<commit>]
                  [-F <file> | -m <msg>] [--reset-author] [--allow-empty]
                  [--allow-empty-message] [--no-verify] [-e] [--author=<author>]
                  [--date=<date>] [--cleanup=<mode>] [--[no-]status]
                  [-i | -o] [--pathspec-from-file=<file> [--pathspec-file-nul]]
                  [(--trailer <token>[(=|:)<value>])...] [-S[<keyid>]]
                  [--] [<pathspec>...]

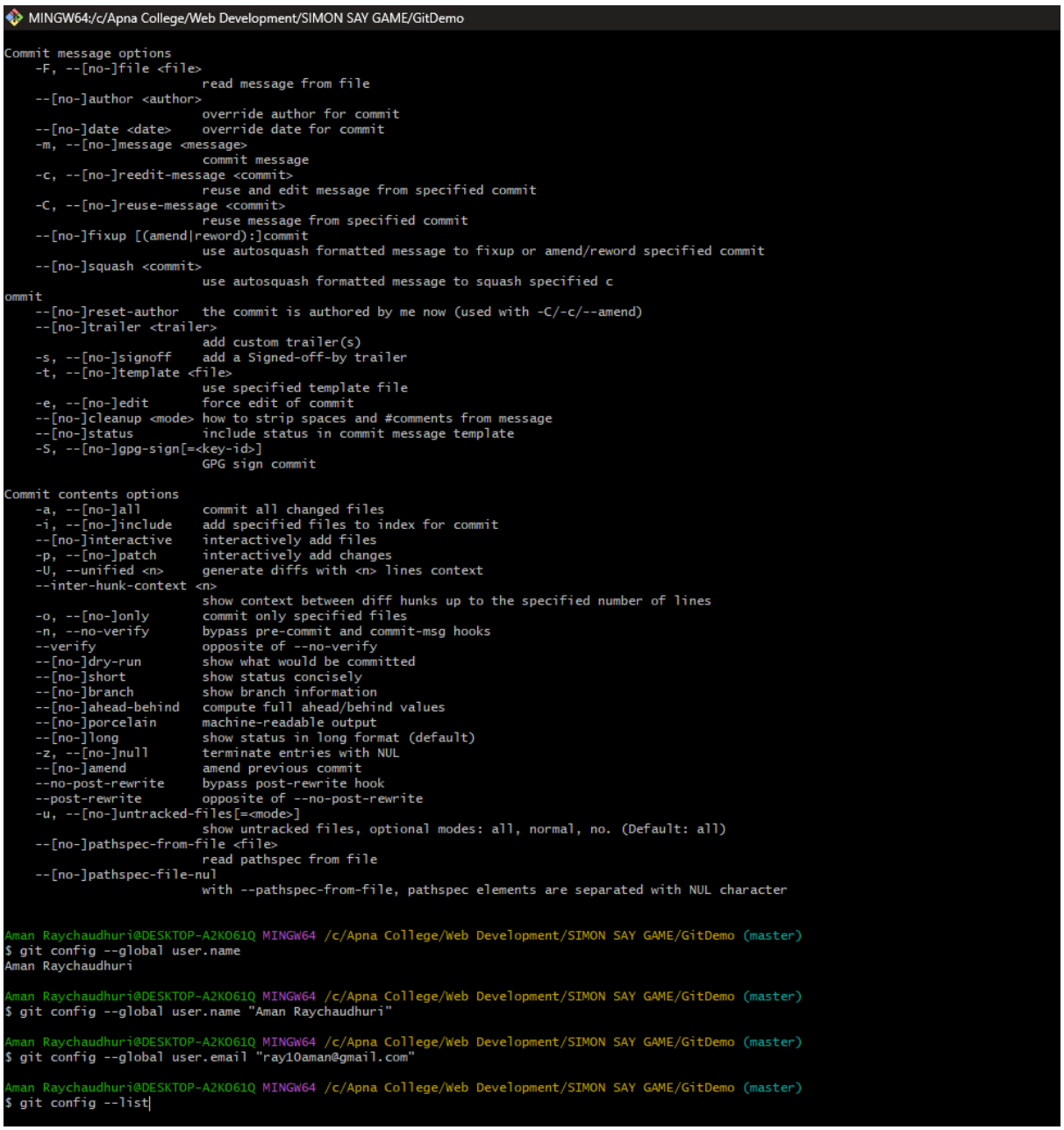
    -q, --[no-]quiet      suppress summary after successful commit
    -v, --[no-]verbose    show diff in commit message template

Commit message options
    -F, --[no-]file <file>
                          read message from file
    --[no-]author <author>
                          override author for commit
    --[no-]date <date>    override date for commit
    -m, --[no-]message <message>
                          commit message
    -c, --[no-]reedit-message <commit>
                          reuse and edit message from specified commit
    -C, --[no-]reuse-message <commit>
                          reuse message from specified commit
    --[no-]fixup [(amend|reword):]commit
                          use autosquash formatted message to fixup or amend/reword specified commit
    --[no-]squash <commit>
                          use autosquash formatted message to squash specified c
ommit
    --[no-]reset-author   the commit is authored by me now (used with -C/-c/--amend)
    --[no-]trailer <trailer>
                          add custom trailer(s)
    -s, --[no-]signoff    add a Signed-off-by trailer
    -t, --[no-]template <file>
                          use specified template file
    -e, --[no-]edit       force edit of commit
    --[no-]cleanup <mode> how to strip spaces and #comments from message
    --[no-]status         include status in commit message template
    -S, --[no-]gpg-sign[=<key-id>]
                          GPG sign commit

Commit contents options
    -a, --[no-]all        commit all changed files
    -i, --[no-]include    add specified files to index for commit
    --[no-]interactive    interactively add files
    -p, --[no-]patch      interactively add changes
    -U, --unified <n>     generate diffs with <n> lines context
    --inter-hunk-context <n>
                          show context between diff hunks up to the specified number of lines
    -o, --[no-]only       commit only specified files
    -n, --no-verify       bypass pre-commit and commit-msg hooks
    --verify              opposite of --no-verify
    --[no-]dry-run        show what would be committed
    --[no-]short          show status concisely
    --[no-]branch         show branch information
    --[no-]ahead-behind   compute full ahead/behind values
    --[no-]porcelain      machine-readable output
    --[no-]long           show status in long format (default)
    -z, --[no-]null       terminate entries with NUL
    --[no-]amend          amend previous commit
    --no-post-rewrite     bypass post-rewrite hook
    --post-rewrite        opposite of --no-post-rewrite
```

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CONCLUSION

The experiment demonstrated the use of essential Git version control commands for repository management, tracking changes, branch handling, and collaboration with GitHub. It provided practical knowledge of managing and maintaining software projects efficiently using Git.